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Granule, preparation containing the granule, production method of the granule, and production method of preparation containing the granule

Abstract

One of the objects of the present invention is to provide a granule capable of sufficiently masking bitterness. Alternatively, one of the objects of the present invention is to provide a preparation containing a granule capable of sufficiently masking bitterness. According to an embodiment of the present invention, a granule is provided including a core substance on which a melt component and an active ingredient are disposed, a gelling substance-containing layer disposed on a surface where the melt component and the active ingredient are disposed, and a hydrophobic polymer-containing layer disposed on a surface of the gelling substance-containing layer.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is based upon and claims the benefit of priority from the Provisional Application No. 63/076,654, filed on Sep. 10, 2020, and the prior Japanese Patent Application No. 2021-146436, filed on Sep. 8, 2021, the entire contents of which are incorporated herein by reference.

FIELD

(1) An embodiment of the present invention relates to a granule for masking the bitterness of an active ingredient or a preparation containing the granule. Alternatively, an embodiment of the present invention relates to a production method of a granule for masking the bitterness of an active ingredient or a production method of the preparation containing the granule.

BACKGROUND

(2) Although many active ingredients which exhibit bitterness are known, the release of the active ingredient in an oral cavity can cause severe discomfort to a patient and significantly reduce medication compliance. The release of the active ingredient from a drug should be inhibited, at least while the drug is present in the oral cavity.

(3) Conventionally, masking of bitterness has been performed by covering an active ingredient-containing layer with a taste masking layer containing a water-insoluble polymer, for example, as described in WO02/096392.

(4) WO2017/146052 describes a pharmaceutical composition particle including a core particle containing a drug, an intermediate layer containing a water-soluble gelling swelling substance to coat an outside of the core particle, and an outer layer containing a water-insoluble substance to coat an outside of the intermediate layer, for the purpose of achieving both shielding of an unpleasant taste and improvement of an elution property.

(5) In WO02/096392 and WO2017/146052, although bitterness is masked by coating the active ingredient-containing composition, in the case of an active ingredient having strong bitterness, it is necessary to thicken a coating layer. In this case, there is a problem whereby the pharmaceutical composition particle after coating becomes large and the feeling of actual taking drug is lost. In addition, in a case where the coating layer is thickened, the coating time becomes longer by that amount, and manufacturing efficiency is degraded. Therefore, there is a need for a preparation which can be easily prepared and which is easy to take while being capable of masking bitterness.

SUMMARY

(6) One of the objects in an embodiment of the present invention is to provide a granule capable of sufficiently masking bitterness. Alternatively, one of the objects in an embodiment of the present invention is to provide a preparation containing the granule capable of sufficiently masking bitterness. Alternatively, one of the objects in an embodiment of the present invention is to provide a production method of the granule capable of sufficiently masking bitterness. Alternatively, one of the objects in an embodiment of the present invention is to provide a production method of the preparation containing the granule capable of sufficiently masking bitterness.

(7) According to an embodiment of the present invention, there is provided a granule including a core substance disposed with a melt component and an active ingredient, a gelling substance-containing layer disposed on a surface where the melt component and the active ingredient are disposed, and a hydrophobic polymer-containing layer disposed on a surface of the gelling substance-containing layer.

(8) The granule may further include a melt component layer containing the melt component and disposed on a surface of the core substance, and an active ingredient-containing layer including the

melt component and the active ingredient may be disposed on the surface of the melt component layer.

(9) The granule may further include an overcoat layer disposed on a surface of the hydrophobic polymer-containing layer.

(10) The gelling substance-containing layer may include a gelling substance having a viscosity of 10 mPa/s or more in a 2% aqueous solution.

(11) The gelling substance-containing layer may include one or more gelling substances selected from a group consisting of sodium carboxymethyl cellulose, polyethylene oxide, sodium polyacrylate, sodium alginate, propylene glycol alginate, xanthan gum, carrageenan, guar gum, tara gum, pectin, hydroxypropyl cellulose, hydroxyethyl cellulose, methyl cellulose, carboxyvinyl polymer, locust bean gum, tamarind seed gum, gum arabic, karaya gum, agar, gelatin, polyvinyl alcohol, and a copolymer with polyvinyl alcohol as a part.

(12) The hydrophobic polymer-containing layer may include one or more hydrophobic polymers selected from ethyl cellulose, acetyl cellulose, cellulose acetate phthalate, carboxymethyl ethyl cellulose, hydroxypropyl methyl cellulose acetate succinate, hydroxypropyl methyl cellulose phthalate, dimethylaminoethyl methacrylate/methyl methacrylate copolymer, methyl acrylate/methacrylate acid copolymer, ethyl acrylate/methyl methacrylate copolymer dispersion, aminoalkyl methacrylate copolymer RS, dried methacrylic acid copolymer LD, aminoalkyl methacrylate copolymer E, methacrylic acid copolymer L, methacrylic acid copolymer S, and polyvinyl acetal diethylaminoacetate.

(13) The content ratio of the gelling substance contained in the gelling substance-containing layer to the hydrophobic polymer contained in the hydrophobic polymer-containing layer may be in a range of 2:3 to 3:8.

(14) According to an embodiment of the present invention, there is provided a preparation including any of the above granules and one or more pharmaceutically acceptable additives.

(15) According to an embodiment of the present invention, there is provided a production method of a granule including disposing a melt component and an active ingredient on a surface of a core substance, forming a gelling substance-containing layer by disposing a gelling substance by dry coating on a surface where the melt component and the active ingredient are disposed, and forming a hydrophobic polymer-containing layer by disposing a hydrophobic polymer on a surface of the gelling substance-containing layer.

(16) A melt component layer may be further formed containing the melt component and disposed on a surface of the core substance, and an active ingredient-containing layer may be formed containing the melt component and the active ingredient on a surface of the melt component layer.

(17) An overcoat layer may be further formed on a surface of the hydrophobic polymer-containing layer.

(18) The gelling substance-containing layer may include a gelling substance having a viscosity of 10 mPa/s or more in a 2% aqueous solution.

(19) The gelling substance-containing layer may include one or more gelling substances selected from a group consisting of sodium carboxymethyl cellulose, polyethylene oxide, sodium polyacrylate, sodium alginate, propylene glycol alginate, xanthan gum, carrageenan, guar gum, tara gum, pectin, hydroxypropyl cellulose, hydroxyethyl cellulose, methyl cellulose, carboxyvinyl polymer, locust bean gum, tamarind seed gum, gum arabic, karaya gum, agar, gelatin, polyvinyl alcohol, and a copolymer with polyvinyl alcohol as a part.

(20) The hydrophobic polymer-containing layer may include one or more hydrophobic polymers selected from ethyl cellulose, acetyl cellulose, cellulose acetate phthalate, carboxymethyl ethyl cellulose, hydroxypropyl methyl cellulose acetate succinate, hydroxypropyl methyl cellulose phthalate, dimethylaminoethyl methacrylate/methyl methacrylate copolymer, methyl acrylate/methacrylic acid copolymer, ethyl acrylate/methyl methacrylate copolymer dispersion, aminoalkyl methacrylate copolymer RS, dried methacrylic acid copolymer LD, aminoalkyl

methacrylate copolymer E, methacrylic acid copolymer L, methacrylic acid copolymer S, and polyvinyl acetal diethylaminoacetate.

(21) The content ratio of the gelling substance contained in the gelling substance-containing layer to the hydrophobic polymer contained in the hydrophobic polymer-containing layer may be in a range of 2:3 to 3:8.

(22) According to an embodiment of the present invention, there is provided a production method of the preparation including mixing a granule produced by the production method described in any of the above and one or more pharmaceutically acceptable additives.

Description

BRIEF DESCRIPTION OF DRAWINGS

(1) FIG. 1 is a schematic view showing a cross-sectional structure of a granule **10** according to an embodiment of the present invention;

(2) FIG. 2 is a schematic view showing a cross-sectional structure of a granule **90** of a comparative example; and

(3) FIG. 3 is a diagram showing a dissolution rate of sitagliptin phosphate in a preparation of an example of the present invention.

DESCRIPTION OF EMBODIMENTS

(4) Hereinafter, a granule according to the present invention and a preparation using the same will be described with reference to the drawings. The granule of the present invention and the preparation using the same are not to be construed as being limited to the description of the embodiments and examples shown below. In the drawings referred to in the present embodiment and the examples to be described later, the same portions or portions having similar functions are denoted by the same symbols, and a repetitive description thereof is omitted.

(5) FIG. 1 is a schematic view (cross-sectional view) showing a granule **10** according to an embodiment of the present invention. The granule **10** includes a core substance **11**, a melt component layer **12** disposed on a surface of the core substance **11**, an active ingredient-containing layer **13** disposed on a surface of the melt component layer **12**, a gelling substance-containing layer **15** disposed on a surface of the active ingredient-containing layer **13**, a hydrophobic polymer-containing layer **17** disposed on a surface of the gelling substance-containing layer **15**, and an overcoat layer **19** disposed on a surface of the hydrophobic polymer-containing layer **17**. In the case where the adhesive property between the core substance **11** and the active ingredient-containing layer **13** is good, the melt component layer **12** may be omitted. The overcoat layer **19** may also be omitted.

(6) The core substance **11** is a carrier for disposing the melt component layer **12**, the active ingredient-containing layer **13**, the gelling substance-containing layer **15**, the hydrophobic polymer-containing layer **17**, and the overcoat layer **19**. The core substance **11** is a substance which serves as a core for disposing the melt component layer **12**, the active ingredient-containing layer **13**, the gelling substance-containing layer **15**, the hydrophobic polymer-containing layer **17**, and the overcoat layer **19** in producing the granule **10**. In order to obtain an adhesive property with the melt component layer **12**, an adsorbent material is used as the core substance **11**. For example, an adsorbent material composed of amberlite IRP-64, ion exchange resin, kaolin, calcium carmellose, hydrous silicon dioxide, magnesium silicate, light anhydrous silicic acid, light liquid paraffin, diatomaceous earth, synthetic aluminum silicate, aluminum oxide, aluminum hydroxide, absorbent cotton, magnesium carbonate, precipitated calcium carbonate, dextrin, silicon dioxide, composite aluminum silicate potassium grain, bentonite, polyethylene fiber, magnesium aluminate metasilicate, and medicinal charcoal, and the like can be used as the core substance **11**.

(7) The core substance **11** is preferably spherical so as to uniformly dispose the melt component

layer **12**, the active ingredient-containing layer **13**, the gelling substance-containing layer **15**, the hydrophobic polymer-containing layer **17**, and the overcoat layer **19**.

(8) The melt component layer **12** is a layer disposed between the core substance **11** and the active ingredient-containing layer **13**. The melt component layer **12** is a base layer for disposing the active ingredient-containing layer **13**. In the case where the active ingredient itself has strong adhesive property to the core substance or the ability of the core substance to carry the active ingredient is high, then the melt component and the active ingredient may be disposed directly on the core substance. On the other hand, by arranging the melt component layer **12** composed of a melt component having a high adhesive property to the core substance **11** on the surface of the core substance **11**, more active ingredients can be adhered to the melt component layer **12**, and the content of the active ingredient in the granule **10** can be effectively increased.

(9) The melt component (first melt component) constituting the melt component layer **12** can be selected from oil-based additives. The first melt component is preferably selected from additives in which contacts with the active ingredient do not denature the active ingredient or exhibit a significant increase in related substances. The melt component layer **12** is preferably formed by a melt layering method, and the first melt component may be selected from additives which are solid at normal temperature. Considering the temperature range generally used in the melt layering method, the first melt component is preferably selected from additives having a melting point of 100° C. or less and is preferably selected from additives having a melting point in a temperature range in which the active ingredient does not denature or exhibit a significant increase in related substances. Additives having such properties are exemplified by glycerin monostearate, macrogol (polyethylene glycol), lauromacrogol, and stearic acid, but not limited thereto.

(10) The melt component layer **12** may be disposed on the surface of the core substance **11** in an amount capable of disposing the active ingredient-containing layer **13** and may be disposed on at least a part of the surface of the core substance **11**. It is preferable that the melt component layer **12** covers 90% or more of the surface of the core substance **11** and preferably covers the entire surface of the core substance **11**. Although there is no particular limitation on the thickness of the melt component layer **12**, it is preferable that the thickness of the melt component layer **12** is as thin as possible from the viewpoint of increasing the active ingredient content per granule **10**. In an embodiment, it is preferable that the first melt component constituting the melt component layer **12** is also disposed in a pore of the core substance **11**. In an embodiment, the interface between the core substance **11** and the melt component layer **12** may have a structure in which the melt component constituting the melt component layer **12** enters from the surface of the core substance **11**. In this case, the core substance **11** and the melt component layer **12** do not need to have a clear interface. By arranging the melt component not only on the surface of the core substance **11** but also in the pore connecting to the surface of the core substance **11**, an anchoring effect on the core substance **11** is imparted to the melt component layer **12**, and the adhesive property of the melt component layer **12** to the core substance **11** is improved.

(11) The active ingredient-containing layer **13** is a layer containing the active ingredient and the melt component and may be disposed on the surface of the core substance **11** or may be disposed via the melt component layer **12**. In the granule **10**, the active ingredient is not particularly limited.

(12) In the case where the melt component layer **12** is disposed on the surface of the core substance **11**, a second melt component is included in the active ingredient-containing layer **13** to bind the active ingredients to each other and to bind the active ingredient to the surface of the melt component layer **12**. The second melt component is preferably selected from additives in which contacts with the active ingredient do not denature the active ingredient or exhibit a significant increase in related substances. Since the active ingredient-containing layer **13** is formed by the melt layering method, the second melt component is selected from additives which are solid at normal temperature. When the temperature range generally used in the melt layering method is considered, the second melt component is preferably selected from additives having a melting point of 100° C.

or less, or a glass transition point of 120° C. or less and is preferably selected from additives having a melting point or a glass transition point in a temperature range in which the active ingredient is not denatured or exhibits a significant increase in related substances. In the case where an additive having a melting point or glass transition point lower than the melting point of the first melting component is selected as the second melting component, when the active ingredient-containing layer **13** is formed by the melt layering method, the active ingredient-containing layer **13** can be disposed on the surface of the melt component layer **12** without significantly affecting the surface structure of the melt component layer **12** or changing the surface structure of the melt component layer **12**. On the other hand, in the case where an additive having a melting point or glass transition point higher than the melting point of the first melt component is selected as the second melt component, when the active ingredient-containing layer **13** is formed by the melt layering method, the surface of the melt component layer **12** is slightly melted, and the interface between the melt component layer **12** and the active ingredient-containing layer **13** is fused, so that the adhesive property of the active ingredient-containing layer **13** to the melt component layer **12** can be improved.

(13) The additive used as the second melting component can be exemplified by stearic acid, glyceryl monostearate, macrogol (polyethylene glycol), carnauba wax, hydrogenated oil, laurmacrogol, palmitic acid, cetyl alcohol, aminoalkyl methacrylate copolymer E, ammonioalkyl methacrylate copolymer, dried methacrylic acid copolymer, hypromellose acetate succinate ester, polyvinylpyrrolidone, and polyvinyl acetal diethyl aminoacetate, and the like, but not limited thereto. The melt component (second melt component) contained in the active ingredient-containing layer **13** may be the same as or different from the melt component (first melt component) contained in the melt component layer **12**.

(14) The active ingredient-containing layer **13** contains the active ingredient as the main component. The active ingredient-containing layer **13** preferably contains 50% by mass or more of the active ingredient relative to the sum of the masses of the active ingredient and the second melt component. In other words, it is preferable that for the active ingredient-containing layer **13** contain a small amount of the second melt component within a range in which the active ingredient-containing layer **13** can be formed on the surface of the melt component layer **12**. Thus, the content of the active ingredient in the granule **10** can be effectively increased.

(15) The gelling substance-containing layer **15** has a function for adjusting the elution of the active ingredient from the active ingredient-containing layer **13**. As will be described later, although the hydrophobic polymer-containing layer **17** suppresses moisture (saliva) in the oral cavity from flowing into the active ingredient-containing layer **13**, the gelling substance contained in the gelling substance-containing layer **15** absorbs moisture soaked from the hydrophobic polymer-containing layer **17** and gelates to form a gel layer. This gel layer can suppress the influx of moisture into the active ingredient containing layer **13** in the oral cavity to prevent elution of the active ingredient from the active ingredient containing layer **13**. In addition, since a large amount of water in the gastric juice dissolves the gel layer, the gelling substance-containing layer **15** can achieve rapid elution of the active ingredient from the active ingredient-containing layer **13** in the stomach.

(16) In this embodiment, a gelling substance having a viscosity of 10 mPa/s or more in a 2% aqueous solution can be used. For example, one gelling substance or a combination of two or more gelling substances selected from sodium carboxy methylcellulose, polyethylene oxide, sodium polyacrylate, sodium alginate, propylene glycol alginate, xanthan gum, carrageenan, guar gum, tara gum, pectin, hydroxypropyl cellulose, hydroxyethyl cellulose, methyl cellulose, carboxyvinyl polymer, locust bean gum, tamarind seed gum, arabic gum, karaya gum, agar, gelatin, polyvinyl alcohol, and a copolymer with polyvinyl alcohol as a part and the like may be used as the gelling substance.

(17) The hydrophobic polymer-containing layer **17** has a function for suppressing moisture in the oral cavity from flowing into the active ingredient-containing layer **13**. The hydrophobic polymer-

containing layer **17** is composed of a water-insoluble substance which is hardly soluble in water or practically insoluble in water. For example, one hydrophobic polymer or a combination of two or more hydrophobic polymers selected from ethyl cellulose, acetyl cellulose, cellulose acetate phthalate, carboxymethyl cellulose, hydroxypropyl methyl cellulose acetate succinate, hydroxypropyl methylcellulose phthalate, dimethylaminoethyl methacrylate/methyl methacrylate copolymer, methyl acrylate/methacrylic acid copolymer, ethyl acrylate/methyl methacrylate copolymer dispersion, aminoalkyl methacrylate copolymer RS, dried methacrylic acid copolymer LD, aminoalkyl methacrylate copolymer E, methacrylic acid copolymer L, methacrylic acid copolymer LD, methacrylic acid copolymer S, and polyvinyl acetal diethylaminoacetate and the like may be used as the water-insoluble substance.

(18) In the granule **10** according to the embodiment, it is possible to sufficiently suppress moisture in the oral cavity from flowing into the active ingredient-containing layer **13** by having the gelling substance-containing layer **15**. Therefore, the hydrophobic polymer-containing layer **17** can be thinned. The time required for producing the granule **10** can be reduced by thinning the hydrophobic polymer-containing layer **17**. In addition, for the granule **10** according to the embodiment, since the gelling substance-containing layer **15** is disposed by the melt layering method, it is possible to greatly reduce the time required for the arrangement of the gelling substance-containing layer **15** as compared with the conventional case.

(19) By arranging the overcoat layer **19** on the surface of the hydrophobic polymer-containing layer **17**, the ingestibility is improved and the elution delay of the active ingredient is suppressed. For example, one or a combination of two or more selected from pregelatinized starch, sodium caseinate, carboxyvinyl polymer, carboxymethyl starch sodium, sucrose fatty acid ester, hydroxypropyl cellulose, hydroxypropyl methyl cellulose, methyl cellulose, hydroxyethyl cellulose, pullulan, polyvinylpyrrolidone, copolyvidone, polyoxyethylene-polyoxypropylene glycol, polyvinyl alcohol-polyethylene glycol graft copolymer, polyvinyl alcohol, macrogol, polyethylene oxide, amino acids such as glycine and alanine, sweeteners such as glycyrrhizic acid, sugars such as dextrin and lactose, sugar alcohols such as mannitol and xylitol, crystalline cellulose, crospovidone, and triethyl citrate and the like may be used for the overcoat layer **19**.

(20) [Production Method of Granule]

(21) The granule **10** according to an embodiment of the present invention can be produced as follows. The core substance **11** and the first melt component are mixed, and the first melt component is disposed on the surface of the core substance **11**. In addition, the first melt component is melted by the melt layering method to form the melt component layer **12** on the surface of the core substance **11**. At this time, the core substance **11** and the first melt component are heated to a temperature equal to or higher than the melting point of the first melt component. Considering the temperature range generally used in the melt layering method, the heating temperature is 100° C. or less. Further, by arranging the first melt component not only on the surface of the core substance **11** but also in the pore connecting to the surface of the core substance **11**, it is preferable to impart an anchoring effect on the core substance **11** to the melt component layer **12** and improve the adhesive property of the melt component layer **12** to the core substance **11**.

(22) The core substance **11** disposed with the melt component layer **12** is mixed with the active ingredient and the second melt component, and the active ingredient and the second melt component are disposed on the surface of the melt component layer **12**. Further, the second melt component is melted by the melt layering method to form the active ingredient-containing layer **13** on the surface of the melt component layer **12**. At this time, the core substance **11** disposed with the melt component layer **12**, the active ingredient, and the second melt component are heated to a temperature equal to or higher than the melting point of the second melt component. Considering the temperature range generally used in the melt layering method, the heating temperature is 100° C. or less.

(23) In an embodiment, in the case where an additive having a melting point or glass transition point lower than the melting point of the first melting component is selected as the second melting component, when the active ingredient-containing layer **13** is formed by the melt layering method, the active ingredient-containing layer **13** can be formed on the surface of the melt component layer **12** without significantly affecting the surface structure of the melt component layer **12** or changing the surface structure of the melt component layer **12** by heating to a temperature equal to or higher than the melting point or glass transition point of the second melting component and lower than the melting point of the first melting component. On the other hand, in the case where an additive having a melting point or glass transition point higher than the melting point of the first melt component is selected as the second melt component, when the active ingredient-containing layer **13** is formed by the melt layering method, the surface of the melt component layer **12** is slightly melted by heating to a temperature higher than the melting point or glass transition point of the second melt component, and the interface between the melt component layer **12** and the active ingredient-containing layer **13** is fused, so that the adhesive property of the active ingredient-containing layer **13** to the melt component layer **12** can be improved. It is preferable to fuse layers in a temperature range that does not denature the active ingredient or exhibit a significant increase in related substances.

(24) A gelling substance is disposed on the surface of the active ingredient-containing layer **13**. In this embodiment, the gelling substance-containing layer **15** is formed by dry coating. In an embodiment, the gelling substance-containing layer **15** can be suitably formed by the melt layering method.

(25) The hydrophobic polymer-containing layer **17** is disposed on the surface of the gelling substance-containing layer **15**. The hydrophobic polymer-containing layer **17** can be formed by coating a hydrophobic polymer on the surface of the gelling substance-containing layer **15** using a mixed solvent of a hydrophilic organic solvent and water. For example, ethanol, isopropanol, or the like can be suitably used as the hydrophilic organic solvent. It is preferable to use a mixed solvent in the range of water:organic solvent=1:2 to 1:100.

(26) The overcoat layer **19** is disposed on the surface of the hydrophobic polymer-containing layer **17**. The overcoat layer **19** can be formed by coating the above-described additive on the surface of the hydrophobic polymer-containing layer **17** using an aqueous solvent such as water or a non-aqueous solvent such as alcohol.

(27) In this embodiment, since the gelling substance-containing layer **15** is formed between the active ingredient-containing layer **13** and the hydrophobic polymer-containing layer **17**, the hydrophobic polymer-containing layer **17** can be thinned. The time required for producing the granule **10** can be reduced by thinning the hydrophobic polymer-containing layer **17**. In the granule **10** according to the embodiment, since the gelling substance-containing layer **15** is disposed by the melt layering method, it is possible to greatly reduce the time required for the arrangement of the gelling substance-containing layer **15** as compared with the conventional case.

(28) [Preparation]

(29) A preparation using the granule **10** can be produced. For example, the granule **10** and one or more pharmaceutically acceptable known additives may be mixed to form a pharmaceutical composition. The pharmaceutical composition may be tableted to form a tablet. The pharmaceutical composition to which a disintegrant is added may be tableted to form an orally disintegrating tablet. The pharmaceutical composition may be encapsulated in a capsule to form a capsule tablet.

(30) [Evaluation of Effect of Suppressing Bitterness]

(31) The effect of suppressing bitterness of the granule **10** according to the present embodiment can be evaluated by a sensory test. The effect of suppressing bitterness of the preparation according to the present embodiment can be evaluated according to a dissolution test method (paddle method) of the Japanese Pharmacopoeia Seventeenth Edition.

Example 1

(32) 256 g of hydrous silicon dioxide (Fuji Silysia Chemical Ltd., Sylopure® P100) was used as the core substance **11**, and 384 g of stearic acid (NOF CORPORATION, Sakura Fine Powder) was used as the first melt component. Hydrous silicon dioxide and stearic acid were poured into a high-speed stirring granulator (Fukae Industrial Co., Ltd., high-speed mixer, FS-GS-5 J), and granulated for 11 minutes at an agitator rotation speed of 300 rpm, a chopper rotation speed of 1500 rpm, and a water temperature of about 80° C. At this time, the temperature of the additive was about 70° C.

(33) 80.0 g of the core substance disposed with the melt component layer **12** on the surface, 248.12 g of sitagliptin phosphate as the active ingredient, and 24.0 g of aminoalkyl methacrylate copolymer E (Evonik Roehm, EUDRAGIT® EPO) as the second melt component were poured into a rotor fluidized bed granulator (Powrex Corporation, Model: MP-01) and granulated at an air supply temperature of 80.0 to 85.0° C. and a rotor rotation speed of 400 rpm. At this time, the temperature of the additive was about 60° C.

(34) 352.12 g of the core substance disposed with the active ingredient-containing layer **13** on the surface, 32.0 g of carboxy vinyl polymer (Lubrizol Advanced Materials, Inc., CARBOPOL® 971P NF), and 6.0 g of sucralose (San-Ei Gen F.F.I., Inc., P) were poured into a rotor fluidized bed granulator (Powrex Corporation, Model: MP-01), and dry coating was performed at an air supply temperature of 80.0 to 85.0° C. and a rotor rotation speed of 400 rpm. At this time, the temperature of the additive was about 60° C. CARBOPOL® 971P NF has a viscosity of 4000 to 11000 mPa/s in a 0.5% aqueous solution.

(35) 48.0 g of dried methacrylic acid copolymer LD (Evonik Roehm, EUDRAGIT® L100-55), 1.44 g of hypromellose (Shin-Etsu Chemical Co., Ltd., TC-5® E), 14.4 g of talc (FUJI TALC INDUSTRIAL CO., LTD., ML115), and 4.8 g of light anhydrous silicic acid (Freund Corporation, Adsolider® 101) were dispersed in 560 g of ethanol and 96.0 g of purified water to prepare a coating liquid. 390.12 g of the core substance disposed with the gelling substance-containing layer **15** on the surface was poured into a rotor fluidized bed granulator (Powrex Corporation, Model: MP-01) and sprayed with a coating agent to form the hydrophobic polymer-containing layer **17**.

(36) 32.0 g of D-mannitol (Mitsubishi Shoji Foodtech Co., Ltd., Mannitol P) and 4.0 g of sucralose (San-Ei Gen F.F.I., Inc., P) were dissolved in 280 g of purified water to prepare a coating liquid. 458.76 g of the core substance disposed with the hydrophobic polymer-containing layer **17** on the surface was poured into a rotor fluidized bed granulator (Powrex Corporation, Model: MP-01) and sprayed with a coating agent to form the overcoat layer **19** to obtain the particle of Example 1.

(37) 140 mg of granules constituting of 113.4 mg of D-Mannitol (Mitsubishi Shoji Foodtech Co., Ltd, Mannitol P), 14.0 mg of crystalline cellulose (Asahi Kasei Corp., CEOLUS® PH-101), 7.0 mg of low substitution hydroxypropyl cellulose (Shin-Etsu Chemical Co., Ltd., L-HPC® NDB-022), 2.8 mg of crospovidone (BASF, Kollidon® CL-F), and 2.8 mg of crospovidone (BASF, Kollidon® CL-M), 247.38 mg of particles of Example 1, 34.62 mg of D-Mannitol (Freund Corporation, Granutol® F), 20.0 mg of silica-treated crystalline cellulose (JRS PHARMA, PROSOLV® SMCC50), 16.0 mg of crospovidone (BASF, Kollidon® CL-F), 10.0 mg of magnesium aluminometasilicate (Fuji Chemical Industries Co., Ltd., Neusilin® UFL-2), and 12.0 mg of sodium stearyl fumarate (Rettenmaier Japan, Co., Ltd., PRUVR) were mixed per tablet in a poly bag, and the mixed powder was tableted at a tableting pressure of 11.3 kN using a rotary tableting press (KIKUSUI SEISAKUSHO LTD., VELA5).

Comparative Example 1

(38) FIG. 2 is a schematic view showing a cross section of a granule **90** of Comparative Example 1. The granule **90** of Comparative Example 1 is different from the granule **10** of Example 1 in that an undercoat layer **95** is disposed on the surface of the active ingredient-containing layer **13** instead of the gelling substance-containing layer **15**.

(39) 32.0 g of hypromellose (Shin-Etsu Chemical Co., Ltd., TC-5® E) and 1.6 g of light anhydrous silicic acid (Freund Corporation, Adsolider® 101) were dispersed in 640 g of purified water to prepare a coating liquid. 352.12 g of the core substance disposed with the active ingredient-

containing layer **13** on the surface obtained by the same method as in Example 1 was poured into a rotor fluidized bed granulator (Powrex Corporation, Model: MP-01) and sprayed with a coating agent to form the undercoat layer **95**.

(40) 80.0 g of dried methacrylic acid copolymer LD (Evonik Roehm, EUDRAGIT® L100-55), 3.2 g of hypromellose (Shin-Etsu Chemical Co., Ltd., TC-5® E), 24.0 g of talc (FUJI TALK INDUSTRIAL CO., LTD., ML115), and 8.0 g of light anhydrous silicic acid (Freund Corporation, Adsolider® 101) were dispersed in 1000 g of ethanol and 200.0 g of purified water to prepare a coating liquid. 390.12 g of the core substance disposed with the undercoat layer **95** on the surface was poured into a rotor fluidized bed granulator (Powrex Corporation, Model: MP-01) and sprayed with a coating agent to form the hydrophobic polymer-containing layer **17**.

(41) The overcoat layer **19** was formed by the same method as in Example 1 to obtain the particle of Comparative Example 1.

(42) 138.68 mg of granules constituting of 103.68 mg of D-mannitol (mannitol P), 12.8 mg of crystalline cellulose (PH-101), 6.4 mg of low-substitution hydroxypropyl cellulose (NDB-022), 2.56 mg of crospovidone (CL-F), and 2.56 mg of crospovidone (CL-M), 278.46 mg of particles of Comparative Example 1, 35.54 mg of D-mannitol (Granutol® F), 16.0 mg of silica-treated crystalline cellulose (SMCC50), 10.0 mg of crospovidone (CL-F), 16.0 mg of magnesium aluminometasilicate (UFL-2), and 16.0 mg of sodium stearyl fumarate (PRUV) were mixed per tablet in a polybag, and the mixed powder was tableted at a tableting pressure of 13.2 kN using a rotary tableting press (KIKUSUI SEISAKUSHO LTD., VELA5).

(43) [Evaluation of Effect of Suppressing Bitterness]

(44) The effect of suppressing bitterness was evaluated by three testers by containing 300 mg of granules of Example 1 or Comparative Example 1 in the mouth of the testers, and the effect of the bitterness masking was evaluated. Although the granule of Example 1 had a lower content of the hydrophobic polymer as compared with the granule of Comparative Example 1, the bitterness of the granule of Example 1 could be suppressed.

(45) Although about 150 minutes was required to form the undercoat layer **95** in Comparative Example 1, the time required to form the gelling substance-containing layer **15** was only about 15 minutes in Example 1. Therefore, it was clarified that the production time of the granule of Example 1 can be greatly reduced.

Example 2

(46) 256 g of hydrous silicon dioxide (Fuji Silysia Chemical Ltd., Sylopure® P100) was used as the core substance **11**, and 384 g of stearic acid (NOF CORPORATION, Sakura Fine Powder) was used as the first melt component. Hydrous silicon dioxide and stearic acid were poured into a high-speed stirring granulator (Fukae Industrial Co., Ltd., high-speed mixer, FS-GS-5 J), and granulated for 11 minutes at an agitator rotation speed of 300 rpm, a chopper rotation speed of 1500 rpm, and a water temperature of about 80° C. At this time, the temperature of the additive was about 70° C.

(47) 80.0 g of the core substance disposed with the melt component layer **12** on the surface, 248.12 g of sitagliptin phosphate as the active ingredient, and 24.0 g of aminoalkyl methacrylate copolymer E (Evonik Roehm, EUDRAGIT® EPO) as the second melt component were poured into a rotor fluidized bed granulator (Powrex Corporation, Model: MP-01), and granulation was performed at an air supply temperature of 80.0 to 85.0° C. and a rotor rotation speed of 400 rpm. At this time, the temperature of the additive was about 60° C.

(48) 352.12 g of the core substance disposed with the active ingredient-containing layer **13** on the surface, 32.0 g of carboxy vinyl polymer (Lubrizol Advanced Materials, Inc., CARBOPOL® 971P NF), and 6.0 g of sucralose (San-Ei Gen F.F.I., Inc., P) were poured into a rotor fluidized bed granulator (Powrex Corporation, Model: MP-01), and dry coating was performed at an air supply temperature of 80.0 to 85.0° C. and a rotor rotation speed of 400 rpm. At this time, the temperature of the additive was about 60° C. The carboxyvinyl polymer has a viscosity of 4000 to 11000 mPa/s in a 0.5% aqueous solution.

(49) 48.0 g of dried methacrylic acid copolymer LD (Evonik Roehm, EUDRAGIT® L100-55), 1.44 g of hypromellose (Shin-Etsu Chemical Co., Ltd., TC-5® E), 14.4 g of talc (FUJI TALC INDUSTRIAL CO., LTD., ML115), and 4.8 g of light anhydrous silicic acid (Freund Corporation, Adsolider® 101) were dispersed in 560 g of ethanol and 96.0 g of purified water to prepare a coating liquid. 390.12 g of the core substance disposed with the gelling substance-containing layer **15** on the surface was poured into a rotor fluidized bed granulator (Powrex Corporation Model: MP-01) and sprayed with a coating agent to form the hydrophobic polymer-containing layer **17**. (50) 32.0 g of D-mannitol (Mitsubishi Shoji Foodtech Co., Ltd., Mannitol P) and 4.0 g of sucralose (San-Ei Gen F.F.I., Inc., P) were dissolved in 280 g of purified water to prepare a coating liquid. 458.76 g of the core substance disposing the hydrophobic polymer-containing layer **17** on the surface was poured into a rotor fluidized bed granulator (Powrex Corporation, Model: MP-01) and sprayed with a coating agent to form the overcoat layer **19** to obtain the particle of Example 2. (51) 140 mg of D-Mannitol-crystalline cellulose-low substitution hydroxypropyl cellulose-crospovidone granules constituting of 113.4 mg of D-Mannitol (Mitsubishi Shoji Foodtech, Co., Ltd, Mannitol P), 14.0 mg of crystalline cellulose (Asahi Kasei Corp., CEOLUS® PH-101), 7.0 mg of low substitution hydroxypropyl cellulose (Shin-Etsu Chemical Co., Ltd., L-HPC® NDB-022), 2.8 mg of crospovidone (BASF, Kollidon® CL-F), and 2.8 mg of crospovidone (BASF, Kollidon® CL-M), 247.38 mg of particles of Example 2, 34.62 mg of D-Mannitol (Freund Corporation, Granutol), 20.0 mg of silica-treated crystalline cellulose (JRS PHARMA, PROSOLV® SMCC50), 10.0 mg of crospovidone (BASF, Kollidon® CL-F), 16.0 mg of magnesium aluminometasilicate (Fuji Chemical Industries Co., Ltd., Neusilin® UFL-2.), and 12.0 mg of sodium stearyl fumarate (Rettenmaier Japan, Co., Ltd., PRUV®) were mixed per tablet in a polybag, and the mixed powder was tableted at a tableting pressure of 11.3 kN using a rotary tableting press (KIKUSUI SEISAKUSHO LTD., VELA5).

Example 3

(52) The gelling substance-containing layer **15** of Example 3 was formed by the same method as in Example 2, except that 352.12 g of the core substance disposed with the active ingredient-containing layer **13** of Example 2 on the surface was used, and carboxy vinyl polymer was changed to 24.0 g and sucralose was changed to 6.2 g.

(53) The hydrophobic polymer-containing layer **17** of Example 3 was formed by the same method as in Example 2 by using 382.52 g of the core substance disposed with the gelling substance-containing layer **15** on the surface.

(54) The overcoat layer **19** of Example 3 was formed by the same method as in Example 2, except that 451.2 g of the core substance disposed with the hydrophobic polymer-containing layer **17** on the surface was used, and sucralose (San-Ei Gen F.F.I., Inc., P) was changed to 4.8 g to obtain the particle of Example 3.

(55) 160 mg of D-Mannitol-crystalline cellulose-low substitution hydroxypropyl cellulose-crospovidone granules, 243.98 mg of particles of Example 3, 29.22 mg of D-Mannitol (Granutol® F), 16.0 mg of silica-treated crystalline cellulose (JRS PHARMA, PROSOLV® SMCC50), 10.0 mg of crospovidone (BASF, Kollidon® CL-F), 16.0 mg of magnesium aluminometasilicate (Fuji Chemical Industries Co., Ltd., Neusilin® UFL-2), and 4.8 mg of sodium stearyl fumarate (Rettenmaier Japan, Co., Ltd., PRUV®) were mixed per tablet in a polybag, and the mixed powder was tableted at a tableting pressure of 11.3 kN using a rotary tableting press (KIKUSUI SEISAKUSHO LTD., VELA5).

Example 4

(56) The gelling substance-containing layer **15** of Example 4 was formed by the same method as in Example 2, except that 352.12 g of the core substance disposed with the active ingredient-containing layer **13** of Example 2 on the surface was used, and carboxy vinyl polymer was changed to 24.0 g, and sucralose was changed to 4.0 g of hydrous silicon dioxide (Evonik, CARPLEX® #80).

(57) 62.0 g of dried methacrylic acid copolymer LD (Evonik Roehm, EUDRAGIT® L100-55), 2.4 g of hypromellose (Shin-Etsu Chemical Co., Ltd., TC-5® E), and 8.0 g of hydrous silicon dioxide (Evonik, CARPLEX® #80) were dispersed in 640 g of ethanol and 160 g of purified water to prepare a coating liquid. 380.12 g of the core substance disposed with the gelling substance-containing layer **15** on the surface was poured into a rotor fluidized bed granulator (Powrex Corporation, Model: MP-01) and sprayed with a coating agent to form the hydrophobic polymer-containing layer **17** of Example 4.

(58) The overcoat layer **19** of Example 3 was formed by the same method as in Example 2, except that 454.6 g of the core substance disposed with the hydrophobic polymer-containing layer **17** on the surface was used and sucralose (San-Ei Gen F.F.I., Inc., P) was changed to 8.0 g to obtain the particle of Example 4.

(59) 176 mg of D-mannitol-crystalline cellulose-low-substitution hydroxypropyl cellulose-crospovidone granules, 247.26 mg of particle of Example 4, 10.74 mg of D-mannitol (Freund Corporation, Granutol® F), 10.0 mg of crospovidone (BASF, Kollidon® CL-F), 24.0 mg of magnesium aluminometasilicate (Fuji Chemical Industries Co., Ltd., Neusilin® UFL-2), and 12.0 mg of sodium stearyl fumarate (Rettenmaier Japan, Co., Ltd., PRUV®) were mixed per tablet in a polybag, and the mixed powder was tableted at a tableting pressure of 11.3 kN using a rotary tableting press (KIKUSUI SEISAKUSHO LTD., VELA5).

Example 5

(60) The gelling substance-containing layer **15** of Example 5 was formed by the same method as in Example 2, except that 352.12 g of the core substance disposed with the active ingredient-containing layer **13** of Example 2 on the surface was used, and the carboxy vinyl polymer was changed to 24.0 g and no sucralose was added.

(61) 80.0 g of dried methacrylic acid copolymer LD (Evonik Roehm, EUDRAGIT® RSPO) were dispersed in 400 g of ethanol and 100.0 g of purified water to prepare a coating liquid. 376.12 g of core substance disposed with the gelling substance-containing layer **15** on the surface was poured into a rotor fluidized bed granulator (Powrex Corporation, Model: MP-01) and sprayed with a coating agent to form the hydrophobic polymer-containing layer **17** of Example 5.

(62) The overcoat layer **19** of Example 5 was formed by the same method as in Example 4 by using 456.2 g of the core substance disposed with the hydrophobic polymer-containing layer **17** on the surface to obtain the particle of Example 5.

(63) 160 mg of D-mannitol-crystalline cellulose-low substitution hydroxypropyl cellulose-crospovidone granules, 248.06 mg of particle of Example 5, 17.14 mg of D-Mannitol (Granutol® F), 17.14 mg of silica-treated crystalline cellulose (JRS PHARMA, PROSOLV® SMCC50), 10.0 mg of crospovidone (BASF, Kollidon® CL-F), 24.0 mg of magnesium aluminometasilicate (Fuji Chemical Industries Co., Ltd., Neusilin® UFL-2), and 4.8 mg of magnesium stearate (Taihei Chemical Industrial Co., Ltd.) were mixed per tablet in a polybag, and the mixed powder was tableted at a tableting pressure of 11.3 kN using a rotary tableting press (KIKUSUI SEISAKUSHO LTD., VELA5).

Comparative Example 2

(64) A core substance disposed with the melt component layer **12** on the surface was obtained by the same method as in Example 2, except that stearic acid (NOF CORPORATION, Sakura fine powder) was changed to stearic acid (NOF CORPORATION, vegetable).

(65) A core substance disposed with the active ingredient-containing layer **13** of Comparative Example 2 on the surface was obtained by the same method as in Example 2, except that a core substance disposed with the melt component layer **12** of Comparative Example 2 on the surface was used.

(66) 32.0 g of hypromellose (Shin-Etsu Chemical Co., Ltd., TC-5® E) and 1.6 g of light anhydrous silicic acid (Freund Corporation, Adsolider® 101) were dispersed in 640 g of purified water to prepare a coating liquid. 352.12 g of the core substance disposed with the active ingredient-

containing layer **13** of Comparative Example 2 on the surface was poured into a rotor fluidized bed granulator (Powrex Corporation, Model: MP-01) and sprayed with a coating agent to form the undercoat layer **95**.

(67) 48.0 g of dried methacrylic acid copolymer LD (Evonik Roehm, EUDRAGIT® L100-55), 1.92 g of hypromellose (Shin-Etsu Chemical Co., Ltd., TC-5® E), 14.2 g of talc (FUJI TALC INDUSTRIAL CO., LTD., ML115), and 4.8 g of light anhydrous silicic acid (Freund Corporation, Adsolider® 101) were dispersed in 600 g of ethanol and 120.0 g of purified water to prepare a coating liquid. 385.72 g of the core substance disposed with the undercoat layer **95** on the surface was poured into a rotor fluidized bed granulator (Powrex Corporation, Model: MP-01) and sprayed with a coating agent to form the hydrophobic polymer-containing layer **17**.

(68) 48.0 g of D-mannitol (Mitsubishi Shoji Foodtech Co., Ltd., Mannitol P) and 8.0 g of sucralose (San-Ei Gen F.F.I., Inc., P) were dissolved in 400 g of purified water to prepare a coating liquid. 454.84 g of the core substance disposed with the hydrophobic polymer-containing layer **17** on the surface was poured into a rotor fluidized bed granulator (Powrex Corporation, Model: MP-01) and sprayed with a coating agent to form the overcoat layer **19** to obtain the particle of Comparative Example 2.

(69) 152 mg of D-Mannitol-crystalline cellulose-low substitution-hydroxypropyl cellulose-crospovidone granules, 255.42 mg of particle of Comparative Example 2, 34.58 mg of D-Mannitol (Freund Corporation, Granutol® F), 16.0 mg of silica-treated crystalline cellulose (JRS PHARMA, PROSOLV® SMCC50), 10.0 mg of crospovidone (BASF, Kollidon® CL-F), 16.0 mg of magnesium aluminometasilicate (Fuji Chemical Industries Co., Ltd., Neusilin® UFL-2), and sodium stearyl fumarate (Rettenmaier Japan, Co., Ltd., PRUV) were mixed per tablet in a polybag, and the mixed powder was tableted at a tableting pressure of 11.3 kN using a rotary tableting press (KIKUSUI SEISAKUSHO LTD., VELA5).

Comparative Example 3

(70) 80.0 g of dried methacrylic acid copolymer LD (Evonik Roehm, EUDRAGIT® L100-55), 3.2 g of hypromellose (Shin-Etsu Chemical Co., Ltd., TC-5® E), 24.0 g of talc (FUJI TALC INDUSTRIAL CO., LTD., ML115), and 8.0 g of light anhydrous silicic acid (Freund Corporation, Adsolider® 101) were dispersed in 1000 g of ethanol and 200.0 g of purified water to prepare a coating liquid. 385.72 g of the core substance disposed with the undercoat layer **95** of Comparative example 2 on the surface was poured into a rotor fluidized bed granulator (Powrex Corporation, Model: MP-01) and sprayed with a coating agent to form the hydrophobic polymer-containing layer **17**.

(71) The overcoat layer **19** was formed by the same method as in Comparative Example 2 by using 500.92 g of the core substance disposed with the hydrophobic polymer-containing layer **17** of Comparative Example 3 on the surface to obtain the particle of Comparative Example 3.

(72) The preparation of Comparative Example 3 was obtained by the same method as in Comparative Example 2, except that D-mannitol-crystalline cellulose-low substitution hydroxypropyl cellulose-crospovidone granules was changed to 128 mg, the particle of Comparative Example 3 was changed to 278.46 mg, and D-mannitol (Freund Corporation, Granutol® F) was changed to 35.54 mg per tablet.

(73) [Evaluation of Effect of Suppressing Bitterness]

(74) The elution property of sitagliptin phosphate were evaluated for the preparations of Examples 2 to 5 and Comparative Examples 2 to 3 in accordance with the dissolution test method (paddle method) of the Japanese Pharmacopoeia Seventeenth Edition. 900 ml of water was used as a test solution. The rotation speed of the paddle was 50 rpm. The dissolution rate of sitagliptin phosphate was measured at 5, 10, 15 and 30 minutes after the start of the test. The measurement results of the dissolution rate of sitagliptin phosphate are shown in FIG. 3. The preparations of Examples 2 to 5 using carbopol which is a gelling substance in the gelling substance-containing layer **15** suppressed elution of sitagliptin phosphate as compared with the preparations of Comparative Examples 2 to 3

having the undercoat layer **95** containing hypromellose, and it was clarified that the carbopol functions in suppressing elution of sitagliptin phosphate (bitterness suppress).

(75) In particular, although the preparations of Examples 2 to 4 have smaller contents of a hydrophobic polymer (Eudragit® L100-55) in the hydrophobic polymer-containing layer **17** than the preparation of Comparative Example 3, the elution of the sitagliptin phosphate is suppressed, and bitterness of sitagliptin phosphate can be suppressed with a small coating amount (reduction in the time for forming the gelling substance-containing layer **15**) by containing the gelling substance in the gelling substance-containing layer **15**. Comparing the preparation of Example 2 and the preparation of Example 5, the same elution behavior of sitagliptin phosphate was exhibited despite the fact that the content of the hydrophobic polymer in the hydrophobic polymer-containing layer **17** was reduced to 60% by weight in the preparation of Example 2. From this result, it was clarified that the suppression of bitterness can be achieved in the preparation according to the present invention when the gelling substance: the hydrophobic polymer=2:3 to 3:8. From the viewpoint of reducing the production time, it is preferable that the content of hydrophobic polymer in the hydrophobic polymer-containing layer **17** is as small as possible.

Example 6

(76) 256 g of hydrous silicon dioxide (Fuji Silysia Chemical Ltd., Sylopure® P100) was used as the core substance **11**, and 384 g of stearic acid (NOF CORPORATION, Sakura Fine Powder) was used as the first melt component. Hydrous silicon dioxide and stearic acid were poured into a high-speed stirring granulator (Fukae Industrial Co., Ltd., high-speed mixer, FS-GS-5 J), and granulated for 11 minutes at an agitator rotation speed of 300 rpm, a chopper rotation speed of 1500 rpm, and a water temperature of about 80° C. At this time, the temperature of the additive was about 70° C.

(77) A core substance disposed with the active ingredient-containing layer **13** on the surface was obtained by the same method as in Example 2 by using 80.0 g of the core substance disposed with the melt component layer **12** on the surface.

(78) 319.70 g of the core substance disposed with the active ingredient-containing layer **13** on the surface and 20.4 g of polyethylene oxide (Dow Chemical, POLYOX® WSR N-60K) were poured into a rotor fluidized bed granulator (Powrex Corporation, Model: MP-01), and dry coating was performed at an air supply temperature of 80.0 to 85.0° C. and a rotor rotation speed of 400 rpm. At this time, the temperature of the additive was about 60° C. Polyethylene oxide has a viscosity of 2000 to 4000 mPa/s in a 2% aqueous solution.

(79) The hydrophobic polymer-containing layer **17** of Example 6 was formed by the same method as in Example 4, except that 390.12 g of the core substance disposed with the gelling substance-containing layer **15** of Example 6 on the surface was used.

(80) The particle of Example 6 was obtained by the same method as in Example 4, except that 458.76 g of the core substance disposed with the hydrophobic polymer-containing layer **17** of Example 6 on the surface was used.

Comparative Example 4

(81) 256 g of hydrous silicon dioxide (Fuji Silysia Chemical Ltd., Sylopure® P100) was used as the core substance **11**, 153.6 g of stearic acid (NOF CORPORATION, vegetable) and 204.8 g of stearic acid (BASF Pharma, Kolliwax® S Fine) were used as the first melt component. Hydrous silicon dioxide and stearic acid were poured into a high-speed stirring granulator (Fukae Industrial Co., Ltd., high-speed mixer, FS-GS-5 J), and granulated for 11 minutes at an agitator rotation speed of 300 rpm, a chopper rotation speed of 1500 rpm, and a water temperature of about 80° C. At this time, the temperature of the additive was about 70° C.

(82) 80.0 g of core substance disposed with the melt component layer **12** on the surface, 210.9 g of sitagliptin phosphate as the active ingredient, and 27.2 g of aminoalkyl methacrylate copolymer E (Evonik Roehm, EUDRAGIT® EPO) as the second melt component were poured into a rotor fluidized bed granulator (Powrex Corporation, Model: MP-01), and granulation was performed at an air supply temperature of 80.0 to 85.0° C. and a rotor rotation speed of 400 rpm. At this time, the

temperature of the additive was about 60° C.

(83) 24.0 g of hypromellose (Shin-Etsu Chemical Co., Ltd., TC-5® E) and 1.2 g of hydrous silicon dioxide (Evonik, CARPLEX® #80) were dispersed in 360 g of purified water to prepare a coating liquid. 376.12 g of the core substance disposed with the active ingredient-containing layer **13** of Comparative Example 2 on the surface was poured into a rotor fluidized bed granulator (Powrex Corporation, Model: MP-01) and sprayed with a coating agent to form the undercoat layer **95**.

(84) 76.0 g of dried methacrylic acid copolymer LD (Evonik Roehm, EUDRAGIT® L100-55), 2.4 g of hypromellose (Shin-Etsu Chemical Co., Ltd., TC-5® E), and 14.2 g of titanium oxide (Toho Titanium, Co., Ltd., NA61) were dispersed in 600 g of ethanol and 120.0 g of purified water to prepare a coating liquid. 401.32 g of the core substance disposed with the undercoat layer **95** on the surface was poured into a rotor fluidized bed granulator (Powrex Corporation, Model: MP-01) and sprayed with a coating agent to form the hydrophobic polymer-containing layer **17**.

(85) 32.0 g of D-mannitol (Mitsubishi Shoji Foodtech Co., Ltd., Mannitol P) and 12.0 g of sucralose (San-Ei Gen F.F.I., Inc., P) were dissolved in 280 g of purified water to prepare a coating liquid. 494.12 g of the core substance disposed with the hydrophobic polymer-containing layer **17** on the surface was poured into a rotor fluidized bed granulator (Powrex Corporation, Model: MP-01) and sprayed with a coating agent to form the overcoat layer **19** to obtain the particle of Comparative Example 4.

(86) [Evaluation of Effect of Suppressing Bitterness]

(87) The effect of suppressing bitterness was evaluated for the particles of Examples 4 and 6 and Comparative Example 4. Specifically, the effect of suppressing bitterness was evaluated by two testers by containing 300 mg of granules of Examples 4 and 6 or Comparative Example 4 in the mouths of the testers, and the effect of suppressing bitterness was evaluated. The results of evaluation of the effect of suppressing bitterness of the particles of Examples 4 and 6 and Comparative Example 4, and the formation times of the gelling substance-containing layer **15** and the undercoat layer **95** are shown in Table 1.

(88) TABLE-US-00001

	Comparative Example 4	Example 6	Example 4	Additive
Carboxyvinyl Polyethylene hypromellose polymer oxide *				
Bitterness	○	○	x	
Layer formation	15 minutes	15 minutes	120 minutes	time

(89) Since the particles of Examples 4 and 6 contained a gelling substance in the gelling substance-containing layer **15**, when contained in the mouth of the testers, the gelling substance swelled by saliva permeating from the gap of the hydrophobic polymer-containing layer **17**, thereby suppressing bitterness derived from the active ingredient. Since the particle of Comparative Example 4 did not contain a gelling substance in the undercoat layer **95**, bitterness could not be suppressed. When comparing the amount of the hydrophobic polymer (Eudragit® L100-55) added in the hydrophobic polymer-containing layer **17** of the particles of Examples 4 and 6 and Comparative Example 4, the amounts were 32 mg and 38 mg, respectively, and it was found that the gelling substance contributed to suppressing bitterness, since Examples 4 and 6 suppressed the bitterness even though the amount of the hydrophobic polymer contained in the particles of Examples 4 and 6 was smaller than that of the particles of Comparative Example 4. From the evaluation result of the effect of suppressing bitterness, not only the production time of the gelling substance-containing layer **15** is reduced, but also the amount of the hydrophobic polymer contained in the hydrophobic polymer-containing layer **17** can be reduced as compared with the conventional technology in the particle according to the present invention. Although the difference between the effect of suppressing bitterness of the particles of Example 4 and the effect of suppressing bitterness of the particles of Example 6 was not observed, in principle, it was shown that any additive which is generally treated as a gelling substance and can be dry coated can be applied to the particle according to the present invention.

(90) An embodiment of the present invention may provide a granule capable of sufficiently masking bitterness. Alternatively, an embodiment of the present invention may provide a

preparation containing the granule capable of sufficiently masking bitterness. Alternatively, an embodiment of the present invention may provide a production method of the granule capable of sufficiently masking bitterness. Alternatively, an embodiment of the present invention may provide a production method of a preparation containing the granule capable of sufficiently masking bitterness.

Claims

1. A granule comprising: a core substance; a melt component layer consisting of a first melt component, the melt component layer disposed on a surface of the core substance; an active ingredient-containing layer containing a second melt component and an active ingredient, the active ingredient-containing layer disposed on a surface of the melt component layer; a gelling substance-containing layer disposed on a surface where the second melt component and the active ingredient are disposed; and a hydrophobic polymer-containing layer disposed on a surface of the gelling substance-containing layer, wherein the core substance is composed of an adsorbent material selected from a group consisting of amberlite IRP-64, ion exchange resin, kaolin, calcium carmellose, hydrous silicon dioxide, magnesium silicate, light anhydrous silicic acid, light liquid paraffin, diatomaceous earth, synthetic aluminum silicate, aluminum oxide, aluminum hydroxide, absorbent cotton, magnesium carbonate, precipitated calcium carbonate, dextrin, silicon dioxide, composite aluminum silicate potassium grain, bentonite, polyethylene fiber, magnesium aluminate metasilicate, and medicinal charcoal, the first melt component contained in the melt component layer is selected from a group consisting of glycerin monostearate, lauromacrogol, and stearic acid, the second melt component in the active ingredient-containing layer is selected from a group consisting of stearic acid, glyceryl monostearate, carnauba wax, hydrogenated oil, lauromacrogol, palmitic acid, cetyl alcohol, aminoalkyl methacrylate copolymer E, ammonioalkyl methacrylate copolymer, dried methacrylic acid copolymer, hypromellose acetate succinate ester, polyvinylpyrrolidone, and polyvinyl acetal diethyl aminoacetate and, wherein the gelling substance-containing layer includes a gelling substance having a viscosity of 10 mPa/s or more in a 2% aqueous solution.
2. The granule according to claim 1, further comprising: an overcoat layer disposed on a surface of the hydrophobic polymer-containing layer.
3. The granule according to claim 1, wherein the gelling substance-containing layer includes one or more gelling substances selected from a group consisting of sodium carboxymethyl cellulose, polyethylene oxide, sodium polyacrylate, sodium alginate, propylene glycol alginate, xanthan gum, carrageenan, guar gum, tara gum, pectin, hydroxypropyl cellulose, hydroxyethyl cellulose, methyl cellulose, carboxyvinyl polymer, locust bean gum, tamarind seed gum, gum arabic, karaya gum, agar, gelatin, polyvinyl alcohol, and a copolymer with polyvinyl alcohol as a part.
4. The granule according to claim 1, wherein the hydrophobic polymer-containing layer includes one or more hydrophobic polymers selected from ethyl cellulose, acetyl cellulose, cellulose acetate phthalate, carboxymethyl ethyl cellulose, hydroxypropyl methyl cellulose acetate succinate, hydroxypropyl methylcellulose phthalate, dimethylaminoethyl methacrylate/methyl methacrylate copolymer, methyl acrylate/methacrylic acid copolymer, ethyl acrylate/methyl methacrylate copolymer dispersion, aminoalkyl methacrylate copolymer RS, dried methacrylic acid copolymer LD, aminoalkyl methacrylate copolymer E, methacrylic acid copolymer L, methacrylic acid copolymer S, and polyvinyl acetal diethylaminoacetate.
5. The granule according to claim 4, wherein the content ratio of the gelling substance contained in the gelling substance-containing layer to the hydrophobic polymer contained in the hydrophobic polymer-containing layer is in a range of 2:3 to 3:8.

6. A preparation comprising: the granule according to claim 1; and one or more pharmaceutically acceptable additives.
